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A High-Precision Machine Learning-Based Prediction Model for Delayed Graft functon(DGF) in Chinese Kidney Transplant Patients: A Multicenter Study

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Article

Keywords: Delayed Graft Function (DGF), Kidney Transplantation, Machine Learning, Multicenter Study, Post-Transplant Outcomes

Posted Date: January 6th, 2025

DOI: https://doi.org/10.21203/rs.3.rs-5617823/v1

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Additional Declarations: There is **NO** Competing Interest.

Abstract

Delayed graft function (DGF) is a severe complication following kidney transplantation, and currently, there is a lack of accurate prediction tools tailored for the Chinese population. This study integrates data from 1,093 kidney transplant cases across four medical centers in China (2016–2024) to develop and validate a machine learning-based model for DGF prediction. By comparing nine machine learning algorithms, we found that the LightGBM model performed best in external validation (AUC = 0.80, accuracy = 0.73). SHAP analysis identified donor GFR, donor hemoglobin, and recipient plasma BNP levels as the primary predictive factors, while also highlighting novel predictors such as donor microscopic hematuria and APTT. Cox regression analysis showed that preoperative dialysis duration in recipients (HR = 1.006, 95% CI: 1.001–1.012) was an independent predictor of DGF recovery. In the follow-up study, we observed that while the DGF mortality group exhibited the most significant kidney function impairment (serum creatinine β = 200.57, eGFR β = -39.91), the prognosis of the DGF survival group was comparable to that of the non-DGF survival group. Additionally, the duration of DGF (16.66 ± 13.73 vs. 15.44 ± 14.62 days) and the number of dialysis treatments (8.13 ± 7.39 vs. 7.78 ± 7.22 sessions) were not significantly associated with prognosis. Based on these findings, we developed an online prediction platform (www.kidney-dgf-match.cn) to support clinical decision-making. This study not only establishes the first high-precision DGF prediction model for the Chinese population but also reveals the potential for favorable outcomes in DGF patients with proper management, offering new insights for optimizing post-transplant management strategies.

1 Introduction

Delayed Graft Function (DGF) is a common and severe complication following kidney transplantation, typically defined as the need for dialysis within the first week post-transplantation [1]. Over the past decade, the incidence of DGF has shown a significant upward trend, reaching 26.3% among adult kidney transplant recipients in 2022 [2]. The occurrence of DGF is influenced by a variety of complex factors, including donor- and recipient-related variables, with its molecular mechanisms primarily involving pathological processes such as ischemia-reperfusion injury and immune responses. Clinically, the most direct and typical symptom of DGF is anuria or oliguria [3]. For patients, DGF not only significantly increases hospitalization costs but also severely undermines their hope of escaping dialysis, heightening anxiety about the success of the transplant. For clinicians, DGF complicates perioperative management, making it more challenging and increasing the risk of graft loss and patient mortality. Even more concerning is the poor prognosis associated with DGF. Numerous studies have shown that DGF significantly reduces long-term graft survival rates and patient survival rates. Moreover, DGF increases the risk of complications such as rejection and infection, subjecting patients to prolonged physical and psychological distress, which can even be life-threatening [4].

Given the significance of DGF, researchers in recent years have proposed several scoring systems to predict and reduce its incidence. For instance, Irish et al. developed a prediction model based on five risk factors for the North American population [5], while Marion Chapal et al. proposed a similar model for

the French population [6]. However, these existing models have limitations, including relatively low accuracy and reliance on single-source data, preventing them from gaining widespread consensus. Notably, as the country with the second-highest number of kidney transplants globally, China still lacks a localized DGF prediction model. This highlights the urgent need to develop a DGF prediction model tailored to the Chinese population to enable accurate prediction of DGF occurrence and duration, guide early clinical interventions, and improve outcomes for kidney transplant patients.

In recent years, machine learning (ML) technology has demonstrated immense potential and applicability in the medical field, particularly in the development of predictive models [7–8]. Machine learning algorithms can efficiently process large-scale, multidimensional medical datasets and identify complex nonlinear relationships, giving them significant advantages in addressing clinical prediction challenges. Various machine learning algorithms, such as random forests, support vector machines, and neural networks, have shown exceptional performance in medical prediction models. These methods not only improve the accuracy of predictive models but also enhance their generalizability, enabling consistent performance across diverse populations. Particularly when utilizing multicenter data, machine learning models exhibit greater stability and applicability, offering new possibilities for the development of personalized medical decision support systems. However, in the field of kidney transplantation, especially in DGF prediction, the application of machine learning methods is still in its infancy, with existing studies limited by small sample sizes and single-center data. Given China's prominent status as a major contributor to global kidney transplantation, developing a machine learning-based DGF prediction model using multicenter data from China is both innovative and necessary.

This study aims to fill existing research gaps and provide more accurate individualized prediction and treatment strategies for kidney transplant patients in China and beyond. We developed an innovative pre-transplant evaluation and matching system, integrating various advanced machine learning algorithms. Drawing on years of accumulated clinical experience, we identified nearly 100 relevant variables encompassing multiple aspects of donors and recipients to comprehensively assess and predict donor-recipient matching. To ensure the robustness and generalizability of the model, we not only utilized data from our center but also collaborated with three other large kidney transplant centers, conducting extensive data training and rigorous external validation. To the best of our knowledge, this is the first pre-transplant matching prediction model based on a large multicenter dataset in China. The development of this model represents a significant breakthrough in this field in China and provides a powerful decision-support tool to improve transplant success rates and patient outcomes. By integrating multicenter data with advanced machine learning techniques, our model is expected to enhance the accuracy of DGF prediction while offering a scientific basis for developing personalized kidney transplantation strategies, thereby potentially optimizing overall transplant outcomes.

2 Materials

2.1 Patients and Inclusion criteria

This study employed a multicenter, large-sample research design, enrolling a total of 1,093 kidney transplant recipients. Data were collected from four major medical centers in China, including the First Affiliated Hospital of China Medical University (366 cases, January 2016 to October 2024), Jilin Hospital (209 cases), Dalian Hospital (211 cases), and the Northern Army General Hospital (307 cases). For the latter three hospitals, the data collection period spanned from March 2021 to October 2024.

To ensure data quality, the following inclusion and exclusion criteria were applied:

1. Donors: Only cases involving deceased donors who were Chinese citizens and met the Chinese national donor standards were included. Cases involving living donors were excluded.
2. Recipients: Cases were excluded if the recipient was under 18 years old, undergoing a second or multiple transplants, involved in dual kidney transplants, or undergoing multi-organ transplants. Cases with key parameter data missing for more than 50% or with a recipient or graft survival time of fewer than 7 days were also excluded.

The study adopted the internationally accepted definition of DGF, which is the requirement for dialysis within the first week post-transplantation. The study protocol was approved by the institutional review board (IRB) of the First Affiliated Hospital of China Medical University and the ethics committees of all participating institutions (IRB No.: AF-SOP-07-1.2-01). As a retrospective study, informed consent from patients was waived. All methods adhered to relevant guidelines and regulations.

2.2 Variables

The primary outcome variable of this study was the occurrence of DGF following kidney transplantation (KTx). A comprehensive set of variables was collected, including recipient-, donor-, graft-, matching-, and transplant immunology-related factors. In addition to baseline characteristics and routine blood and urine laboratory tests, the following parameters were specifically included:

- • Donor-Related Variables: Donor Blood Type, Donor Hypertension (Doner Hp), Donor Infect, Kidney Donor Profile Index (KDPI), Kidney Donor Risk Index (KDRI).
- • Graft-Related Variables: Renal Artery Variation, Renal Vein Variation, Cold ischemia time (CIT).
- • Recipient-Related Variables: Recipient Blood Type, Recipient Nephropathy Etiology, Recipient Preoperative Dialysis Modality, Recipient Preoperative Systolic Blood Pressure, Recipient Preoperative Diastolic Blood Pressure, Recipient Preoperative Pulse Pressure, Recipient Preoperative mean arterial pressure, Recipient Preoperative Urine.
- • Immunology-Related Variables: Immunoinduction Protocols, HLA Mismatches (HLA MM), Panel reactive antibody levels (PRA).
- • Matching-Related Variables: ABO mismatch (ABO MM), Sex mismatch (Sex MM), Age mismatch (Age MM), BMI ratio (BMI Ratio), Deviation of BMI ratio from 1 (BMI Ratio to 1), Body surface area ratio (BSA Ratio), Deviation of BSA ratio from 1 (BSA Ratio to 1).

2.3 Feature Engineering

In multicenter kidney transplantation studies, data missingness is a common challenge due to differences in management practices and organ sources from various locations. To address this issue, we carefully evaluated multiple approaches for handling missing data, including traditional methods such as mean or median imputation, as well as more advanced techniques like IterativeImputer and KNNImputer [9–10]. Considering that certain laboratory values might be entirely absent in specific centers, we opted for IterativeImputer and KNNImputer to ensure model accuracy while minimizing bias and the risk of overfitting.

The IterativeImputer method is advantageous in its ability to capture complex relationships between features. Through an iterative process, the method refines imputation results step by step, making it particularly suitable for clinical datasets with intricate interdependencies. This approach simulates the clinical decision-making process where physicians consider multiple related factors, thereby preserving the internal logical structure of the data.

On the other hand, the KNNImputer method leverages the K-nearest neighbors algorithm, estimating and imputing missing values based on the information from the K most similar neighbors. This method works particularly well for clinical data exhibiting local similarities, effectively capturing subtle yet significant variations within patient populations.

2.4 Data Preprocessing

Given that the incidence of DGF across the participating centers was consistently around 20%, we first addressed the issue of class imbalance in the dataset. Using the RandomUnderSampler, we undersampled the majority class to balance the proportions of positive and negative classes [11]. This step was crucial for preventing the model from being biased toward the majority class, thereby improving its ability to predict the minority class.We also applied the StandardScaler to standardize the features, ensuring that all variables were on the same scale. Standardization is critical for enhancing the performance of many machine learning algorithms by preventing features with larger ranges from dominating those with smaller ranges.

For model selection, we compared the performance of multiple machine learning algorithms, including: Logistic Regression (LR), Gaussian Naive Bayes (GNB), Decision Tree, Random Forest (RF), Support Vector Machine (SVM), K-Nearest Neighbors (KNN), Neural Network, XGBoost, and LightGBM. This multi-model comparison strategy allowed us to comprehensively evaluate the performance of different algorithms on this specific task [12–21].

For Logistic Regression and Random Forest, we further applied Bayesian optimization (via BayesSearchCV) to fine-tune hyperparameters. Compared to traditional grid search or random search methods, Bayesian optimization is more efficient and can identify optimal hyperparameter combinations in a shorter amount of time. This helped enhance the performance of the selected models while reducing computational overhead.

2.5 Model Evaluation

To account for regional population differences, we divided the dataset from the First Affiliated Hospital of China Medical University, Dalian Hospital, and Jilin Hospital into an internal dataset for model training and testing, while using data from the remaining hospital as an external test set to validate the model.For the internal dataset, we initially employed a 70% training set and 30% test set split to evaluate model performance. The evaluation metrics included the area under the receiver operating characteristic curve (ROC-AUC) and accuracy.

To further ensure a comprehensive and robust assessment of each model's performance, we implemented a 10-fold cross-validation approach. This method not only maximizes the use of limited data but also effectively reduces the risk of overfitting, providing a more reliable estimate of model performance. Using the StratifiedKFold method, we randomly divided the dataset into 10 equal-sized subsets. For each model, we performed 10 training and testing cycles, where 9 subsets were used as the training set and the remaining 1 subset served as the validation set.

For performance evaluation during cross-validation, we calculated multiple metrics [22], including Accuracy, Precision, Recall, F1 Score, Specificity, ROC-AUC, and Precision-Recall AUC (PR-AUC). After completing the cross-validation, we selected the model that demonstrated the best performance based on these metrics for external test set validation and final performance evaluation.

2.6 Cox Proportional Hazards Regression Analysis

To assess the impact of various factors on the time to DGF recovery, we conducted a Cox proportional hazards regression analysis. First, we performed univariate Cox regression analysis on all variables to identify potential significant predictors. Using the survival package in R, we analyzed DGF recovery time as the outcome variable and the DGF status as the event indicator. For each variable, we calculated Hazard Ratio (HR), 95% Confidence Interval (CI) and p-value.

Variables with a p-value < 0.1 in the univariate analysis were considered potential significant predictors and were included in the subsequent multivariate Cox regression analysis. The multivariate analysis allowed us to determine the independent effects of these significant predictors on DGF recovery time while adjusting for the influence of other variables. Finally, we visualized the results, providing insights into the key predictors and their relative contributions to DGF recovery.

2.7 Follow-up Analysis

This study conducted long-term follow-up of patients from the transplantation center at the First Affiliated Hospital of China Medical University, concluding in October 2024, using a combination of outpatient follow-ups and telephone interviews, with patients considered lost to follow-up if they could not be contacted after three consecutive phone attempts or if they voluntarily withdrew from the study. Patients were categorized into four groups based on their DGF status and prognosis: DGF Death Group, DGF Survival Group, Non-DGF Death Group, and Non-DGF Survival Group. The primary endpoints included overall patient survival rate, graft survival rate, and mortality rate, while secondary endpoints focused on the longitudinal changes in renal function indicators, including serum creatinine, eGFR, blood urea, and

cystatin C, as well as the incidence of clinical complications. Follow-up data were collected at postoperative intervals of 1, 3, 6, 12, 24, 36, and 48 months. Longitudinal data analysis was performed using the Generalized Estimating Equations (GEE) method with an exchangeable correlation structure, using the Non-DGF Survival Group as the control group, as this method accounts for the correlation between repeated measurements within individuals, is robust to missing data, and incorporates both time effects and group effects [23]. The impact of DGF status and prognosis on renal function trends was assessed using coefficient estimates, confidence intervals, and model fit indices such as the Quasi-likelihood under the Independence Model Criterion (QIC), enabling a robust evaluation of how DGF status and prognosis influence long-term renal function and complication occurrences.

2.8 Website Visualization

To enhance the clinical applicability and accessibility of our predictive models, we developed an interactive web-based visualization tool. This tool, accessible at http://www.kidney-dgf-match.cn/, allows healthcare professionals to input patient-specific data and obtain real-time predictions for DGF risk. The interface was designed to be user-friendly, featuring separate sections for donor characteristics, recipient characteristics, organ preservation details, and immunology-related factors. Users can input a wide range of variables, including demographic information, laboratory test results, and transplant-specific parameters. The tool utilizes best models to provide predictions, offering a comparative view of the results from these two high-performing algorithms. This web-based tool not only serves as a practical application of our research findings but also as a potential decision support system for clinicians involved in kidney transplantation.

2.9 Statistical analysis

All our data can be classified as continuous and categorical variables. We use Shapiro–Wilk test to detect normal distribution in continuous variables, the two independent samples t-test for the normal distribution, and the Mann-Whitney U test for unnormal distribution. In categorical variables, Chi-Squared Test is used to calculate statistical significance. If P-value < 0.05, we consider it remains statistical significance. All model weights are visualized through the SHapley Additive exPlanations (SHAP) [24]. All methods are generated in Jupiter-notebook with kernel Python 3.9 with the scikit package [25].

3 Result

3.1 Basic clinical characteristics comparation

According to Table 1, this study compared the baseline clinical characteristics of kidney transplant recipients with DGF (292 cases) and without DGF (801 cases). Regarding donor characteristics, there were no significant differences between the two groups in terms of age, sex, height, weight, BMI, and body surface area (BSA) ($P > 0.05$). However, significant differences were observed in donor blood type ($P < 0.001$), infection status ($P < 0.001$), and renal artery variation ($P < 0.001$). Specifically, the DGF group

had a higher rate of donor infections (12.3% vs. 3.5%) but a lower incidence of renal artery variation (1.8% vs. 9.6%).

For recipient characteristics, no significant differences were found between the two groups in blood type distribution, sex, height, weight, BMI, or urine output ($P > 0.05$). However, notable differences were identified in ABO matching ($P < 0.001$), age matching ($P < 0.001$), preoperative systolic blood pressure ($P < 0.001$), pulse pressure ($P = 0.001$), and mean arterial pressure ($P = 0.001$), with the DGF group showing lower preoperative systolic and mean arterial pressures.

In terms of immunological features, the DGF group had a higher number of HLA mismatches ($P < 0.001$), a higher rate of PRA positivity (12.3% vs. 7.0%, $P = 0.007$), and was more likely to receive ATG as the induction therapy (77.2% vs. 83.1%, $P < 0.001$). Additionally, there were significant differences in the etiology of kidney disease ($P < 0.001$); glomerulonephritis was more prevalent in the DGF group (28.1% vs. 24.0%), while hypertensive nephropathy and diabetic nephropathy were more common in the non-DGF group.

These findings suggest that certain donor and recipient characteristics, such as donor infection, HLA mismatches, and preoperative blood pressure, may be associated with the occurrence of DGF, providing important clues for further analysis.

Table 1
Baseline characteristics of kidney transplant recipients with and without DGF.

	DGF(n = 292)	Non-DGF(n = 801)	P value
Donor			
Donor Age, median ± SD	54.0 ± 10.2	53 ± 11.2	0.844
Donor Sex, n(%)			0.971
Male	248.0(85.1)	677(84.5)	
Female	44.0(14.9)	124(15.5)	
Donor Height_(cm), median ± SD	170.0 ± 5.4	170 ± 6.9	0.539
Donor Weight_(kg), median ± SD	70.0 ± 10.7	73 ± 15.3	0.237
DBMI, median ± SD	24.0 ± 3.3	24.4 ± 4.3	0.268
DBSA, median ± SD	1.8 ± 0.2	1.8 ± 0.2	0.314
Donor Blood Type, n(%)			< 0.001
O	92.0(31.6)	268.0(33.5)	
A	74.0(25.4)	205.0(25.6)	
B	105.0(36.0)	261.0(32.6)	
AB	21.0(7.0)	67.0(8.3)	
Donor Hp, n(%)			0.235
N	131.0(44.7)	307.0(38.3)	
Y	161.0(55.3)	494.0(61.7)	
Donor infect, n(%)			< 0.001
N	256.0(87.7)	773.0(96.5)	
Treponema	31.0(10.5)	13.0(1.6)	
HBV	0.0(0.0)	0.0(0.0)	
HCV	0.0(0.0)	0.0(0.0)	
Coinfection	5.0(1.8)	15.0(1.9)	
KDPI, median ± SD	66.0 ± 18.0	64.0 ± 20.3	0.530
KDRI, median ± SD	1.2 ± 0.3	1.2 ± 0.3	0.795
Renal Artery Variation, n(%)			0.000

	DGF(n = 292)	Non-DGF(n = 801)	P value
N	287.0(98.2)	724.0(90.4)	
Y	5.0(1.8)	77.0(9.6)	
Renal Vein Variation, n(%)			0.147
N	292.0(100.0)	793.0(99.0)	
Y	0.0(0.0)	8.0(1.0)	
Recipient			
CIT, median ± SD	7.6 ± 3.3	7.0 ± 3.9	0.592
Blood Type, n(%)			0.469
O	84.0(28.9)	269.0(33.6)	
A	74.0(25.4)	200.0(24.9)	
B	105.0(36.0)	266.0(33.2)	
AB	29.0(9.7)	66.0(8.3)	
ABO MM, n(%)			< 0.001
ABO-identical	277.0(94.7)	745.0(93.0)	
ABO-compatible	15.0(5.3)	56.0(7.0)	
Recipient Sex, n(%)			0.625
Male	205.0(70.2)	586.0(73.2)	
Female	87.0(29.8)	215.0(26.8)	
Sex MM, n(%)			0.646
Male to male	187.0(64.0)	512.0(63.9)	
Male to female	61.0(21.1)	163.0(20.4)	
Female to male	18.0(6.1)	75.0(9.3)	
Female to female	26.0(8.8)	51.0(6.4)	
Recipient Age, median ± SD	46.0 ± 11.1	46 ± 10.3	0.797
Age MM, n(%)			< 0.001
>50 to >50	49.0(16.7)	259.0(32.3)	
>50 to <50	156.0(53.5)	225.0(28.1)	
<50 to >50	36.0(12.3)	79.0(9.9)	

	DGF(n = 292)	Non-DGF(n = 801)	P value
<50 to <50	51.0(17.5)	238.0(29.7)	
Height_(cm), median ± SD	171.0 ± 8.0	172 ± 8.0	0.481
Weight_(kg), median ± SD	68.0 ± 15.1	69 ± 13.3	0.630
BMI, median ± SD	23.2 ± 4.02	23.6 ± 3.6	0.547
BMIr, median ± SD	0.9 ± 0.2	1.0 ± 0.2	0.171
BMIr to 1, n(%)			0.089
N	159.0(54.4)	360.0(45.0)	
Y	133.0(45.6)	441.0(55.0)	
RBSA, median ± SD	1.8 ± 0.2	1.8 ± 0.2	0.571
BSAr, median ± SD	1.0 ± 0.1	1.0 ± 0.2	0.181
BSAr to 1, n(%)			0.007
N	84.0(28.9)	405.0(50.5)	
Y	208.0(71.1)	396.0(49.5)	
RNE, n(%)			< 0.001
Glomerulonephritis	82.0(28.1)	192.0(24.0)	
Hypertensive nephropathy	10.0(3.5)	54.0(6.7)	
Diabetic nephropathy	28.0(9.6)	54.0(6.7)	
Polycystic kidney	8.0(2.6)	30.0(3.8)	
Unknown	164.0(56.2)	471.0(58.8)	
Pdm, n(%)			0.135
N	3.0(0.9)	28.0(3.5)	
Hemodialysis	261.0(89.4)	658.0(82.1)	
Peritoneal dialysis	28.0(9.7)	115.0(14.4)	
PSBP, median ± SD	144.0 ± 21.4	152 ± 21.2	< 0.001
PDP, median ± SD	89 ± 13.0	90 ± 13.8	0.029
PSDP, median ± SD	54.0 ± 15.9	60 ± 15.2	0.001
PMAP, median ± SD	116 ± 16.2	120 ± 16.7	0.001
RPU, n(%)			0.312

	DGF(n = 292)	Non-DGF(n = 801)	P value
N	230.0(78.9)	578.0(72.2)	
Y	62.0(21.1)	223.0(27.8)	
HLA MM, n(%)			< 0.001
≤ 3	74.0(25.5)	361.0(45.1)	
>4	218.0(74.5)	440.0(54.9)	
PRA, n(%)			0.007
Negative	256.0(87.7)	745.0(93.0)	
Positive	36.0(12.3)	56.0(7.0)	
Immunoinduction protocols, n(%)			< 0.001
Basiliximab	67.0(22.8)	135.0(16.9)	
ATG	225.0(77.2)	666.0(83.1)	

3.2 Visualization of Model Performance

We divided the training dataset into a 7:3 ratio for model development, and as shown in Fig. 2, the Random Forest and LightGBM models demonstrated the best performance with an AUC of 0.79, followed closely by the XGBoost model with an AUC of 0.76. However, in terms of accuracy, the LightGBM model outperformed all other models, achieving an accuracy of 80%, followed by XGBoost and Random Forest, both with an accuracy of 79%.

Considering that a single train-test split may introduce randomness and that the Random Forest model is prone to overfitting, potentially limiting its generalization ability, we further employed a 10-fold cross-validation method to evaluate model performance. Figure 3 illustrates the results of the 10-fold cross-validation, revealing the consistently strong and stable performance of the Random Forest, XGBoost, and LightGBM models across multiple validation rounds.

Table 2 provides detailed performance metrics for each model during 10-fold cross-validation. Notably, the Random Forest, LightGBM, and XGBoost models achieved significantly higher average accuracy and AUC compared to other models, showcasing their robust predictive power and stability.

3.3 External Validation Results

After performing 10-fold cross-validation on the training set and selecting the top three models, we further validated the generalization ability of the Random Forest, XGBoost, and LightGBM models using an external dataset.

As shown in the ROC curves and performance metrics (Fig. 4 & Table 3), the LightGBM model demonstrated the best performance during external validation. It achieved the highest scores across all evaluation metrics, with an ROC-AUC of 0.80, accuracy of 0.73, precision of 0.69, recall of 0.35, F1 score of 0.47, and specificity of 0.92. These results indicate that the LightGBM model performed exceptionally well in distinguishing between positive and negative samples, showcasing strong generalization ability.

The XGBoost model followed closely behind, delivering solid performance with an ROC-AUC of 0.73 and accuracy of 0.71. While it outperformed the Random Forest model across most metrics, it fell slightly short of the LightGBM model. In contrast, the Random Forest model showed relatively weaker performance on the external dataset. Its ROC AUC was 0.63, and its accuracy was 0.67, with noticeably lower precision, recall, and F1 scores compared to the other two models. However, it is worth noting that the Random Forest model achieved a high specificity of 0.98, indicating strong performance in identifying negative samples.

3.4 Feature Importance Visualization

To better understand the decision-making process of the models and enhance their interpretability, we performed SHAP (SHapley Additive exPlanations) analysis on the top-performing XGBoost and LightGBM models. SHAP values quantify the contribution of each feature to the model's predictions, providing insights into the key factors driving DGF prediction.

Figure 5 presents the SHAP summary plot for the LightGBM model, clearly highlighting the top five features influencing DGF prediction: pBNP, DeGFR, DHb, DuEC, and DuRBC. The analysis revealed that lower DeGFR values significantly increased the risk of DGF, which aligns with clinical understanding, as decreased glomerular filtration rate typically reflects poorer kidney function. Additionally, higher pBNP levels were associated with an increased risk of DGF, potentially reflecting the impact of underlying cardiovascular issues on transplant outcomes. Interestingly, higher DHb values and mild abnormalities in routine urine tests, such as elevated DuEC and DuRBC, also showed a trend of increased DGF risk.

Figure 6 displays the SHAP summary plot for the XGBoost model, which further corroborates the importance of several key features. Similar to the LightGBM model, DeGFR, DHb, and pBNP were confirmed as critical predictors. Additionally, the XGBoost model identified DAPPT and pBUN as important features. Notably, lower DAPPT values were associated with an increased risk of DGF, offering a new perspective on the role of donor and recipient coagulation status in transplant outcomes.

The consistency between the two models in identifying major predictors, particularly DeGFR, DHb, and pBNP, strengthens confidence in the significance of these features for DGF prediction. At the same time, the unique features identified by each model, such as DAPPT and pBUN, provide diverse perspectives for risk assessment, helping to capture the intricate network of factors influencing DGF.

3.5 DGF Recovery Time

Univariate Cox regression analysis (Fig. 7A) identified several factors associated with DGF recovery time ($p < 0.1$). Among these, donor type (HR = 1.66, 95% CI: 1.233–2.236, p = 0.0008), recipient serum creatinine level (HR = 1.002, 95% CI: 1.000–1.003, p = 0.0044), and donor serum calcium level (HR = 0.246, 95% CI: 0.088–0.685, p = 0.0073) had statistically significant impacts on recovery time. Other factors with significant associations included recipient weight, primary kidney disease type, recipient BMI, body surface area, dialysis duration, and donor serum magnesium levels.

In multivariate Cox regression analysis (Fig. 7B), two independent predictors were strongly associated with DGF recovery time after stepwise selection. First, dialysis duration (Pdt) showed statistical significance (HR = 1.006, 95% CI: 1.001–1.012, p = 0.021), indicating that each additional unit of dialysis time increased the risk of prolonged DGF recovery by 0.6%. Second, recipient preoperative systolic blood pressure (PsBp) exhibited marginal significance (HR = 0.094, 95% CI: 0.998–1.032, p = 0.094), suggesting that it may have a modest influence on DGF recovery time, though its p-value slightly exceeded the traditional threshold of 0.05.

3.6 Long-Term Prognosis of DGF

Longitudinal analysis of renal function parameters over time (Fig. 8) revealed distinct trends among the four patient groups. The Non-DGF Survival Group showed the most stable renal function, with a steady increase in eGFR (from 60.90 to 74.89 mL/min/1.73m²) and consistently low serum creatinine levels (101.00–133.30 μmol/L). In contrast, the DGF Survival Group exhibited poorer early renal function but gradual improvement over time, with serum creatinine decreasing from 255.26 to 122.93 μmol/L, although a significant increase was observed at 36 months (289.25 μmol/L). The Death Groups (both DGF Death and Non-DGF Death) showed highly variable and generally worse trends, particularly in the DGF Death Group, where serum creatinine and urea levels were markedly elevated in the early stages.

To quantify the differences among groups, a GEE model was applied using the Non-DGF Survival Group as the reference (Table 3). The analysis showed that the DGF Death Group exhibited the most severe renal dysfunction, with significantly higher serum creatinine (β = 200.57, 95% CI: 78.30–322.84, p = 0.001), significantly lower eGFR (β = -39.91, 95% CI: -47.61 to -32.21, $p < 0.001$), and significantly elevated levels of urea (β = 13.51, 95% CI: 7.40–19.61, $p < 0.001$) and cystatin C (β = 1.64, 95% CI: 0.92–2.36, $p < 0.001$). The Non-DGF Death Group showed significant elevation only in urea levels (β = 8.89, 95% CI: 2.83–14.94, p = 0.004). Meanwhile, the DGF Survival Group exhibited no statistically significant differences in most renal function parameters compared to the control group, except for a slight increase in cystatin C levels (β = 0.58, 95% CI: 0.09–1.06, p = 0.020).

Time effect analysis indicated that urea (β = -0.11, $p < 0.001$) and cystatin C (β = -0.02, p = 0.020) levels decreased significantly over time, suggesting that renal function in surviving patients gradually improved. These findings suggest that while DGF patients experience slower early recovery, their long-term prognosis in terms of renal function is similar to that of Non-DGF patients, provided they survive the early stages.

Table 4
Comparison of Dialysis-related Characteristics Between DGF Survival and Death Groups.

Variable	DGF Survival (n = 38)	DGF Death (n = 9)	P value
DGF Duration (days)	16.66 ± 13.73	15.44 ± 14.62	0.735
Number of Dialysis	8.13 ± 7.39	7.78 ± 7.22	0.807
Mean Dialysis Interval (days)	1.89 ± 0.58	1.75 ± 0.70	0.649

3.7 Interactive Web-Based Tool

To enhance the clinical utility and accessibility of our predictive model, we developed an interactive web-based visualization tool (http://www.kidney-dgf-match.cn/). The website features an intuitive and user-friendly interface that organizes the predictive variables into multiple modules, including donor characteristics, recipient characteristics, organ preservation, and immunology (Fig. 9).

Clinicians can input patient demographic information, laboratory test results, and transplant-related parameters, and the system will compute the risk probability of DGF in real-time. The tool incorporates the significant predictive variables identified in this study, with their respective weight coefficients integrated into the calculation model, providing transplant physicians with a convenient decision support tool.

This online platform represents a practical application of our research findings and offers healthcare professionals in the field of kidney transplantation an objective risk assessment method. By enabling personalized perioperative management strategies, it aims to improve patient outcomes and optimize transplant success rates.

4 Discussion

Among all the machine learning models evaluated, the ensemble learning models XGBoost and LightGBM demonstrated outstanding predictive performance on the external validation dataset, which can be attributed to their unique algorithmic features [26]. XGBoost (eXtreme Gradient Boosting) incorporates a second-order Taylor expansion to approximate the objective function and includes a regularization term to control model complexity, enhancing its robustness in handling nonlinear relationships. Furthermore, its innovative split-point finding algorithm efficiently processes sparse data, making it particularly advantageous in addressing the prevalent issue of missing values in medical datasets. In contrast, LightGBM's standout features include the "Gradient-based One-Side Sampling" (GOSS) and "Exclusive Feature Bundling" (EFB) strategies. GOSS retains samples with large gradients while randomly sampling those with small gradients, significantly reducing computational complexity while maintaining training accuracy. This was particularly effective in handling our imbalanced dataset, as DGF is a relatively rare clinical event. Additionally, LightGBM's leaf-wise growth strategy, which prioritizes optimal splits, allowed the model to better capture the complex nonlinear relationships in

medical data, resulting in superior predictive performance on the external validation set (ROC-AUC = 0.80, accuracy = 0.73).

In terms of data preprocessing, the IterativeImputer outperformed KNNImputer in both prediction accuracy and model robustness. This may be due to the inherent limitations of KNNImputer, which relies on finding the most similar neighbors. As data dimensions increase, the definition of similarity becomes less reliable, and the heterogeneity of populations in kidney transplantation amplifies the errors introduced by this method. Conversely, IterativeImputer focuses on the complex interrelationships between features, leveraging these connections to handle missing values more effectively, which resulted in better outcomes. Thus, for large-scale medical machine learning, IterativeImputer appears to be a more suitable choice.

SHAP visualizations of the two models identified several groups of features with significant predictive value for DGF, spanning organ function, blood status, and immunological factors. Among donor-related factors, DeGFR was the most significant predictor, with lower donor GFR values substantially increasing the risk of DGF. This finding aligns with clinical experience, as poor baseline kidney function directly impacts early graft recovery [27]. Donor hemoglobin (DHb) levels showed a complex bidirectional influence: moderately elevated Hb levels may improve oxygen delivery to tissues, but excessively high Hb levels might indicate hemoconcentration, inadequate fluid maintenance, or poor organ perfusion, increasing the risk of perfusion-related injury. This highlights the importance of precise management of donor hemoglobin and fluid levels to optimize organ perfusion quality during the maintenance period. Additionally, recipient preoperative blood urea nitrogen (pBUN) levels showed significant predictive value, likely reflecting the impact of poor preoperative dialysis quality, metabolic imbalances, and nitrogen retention [28].

In terms of cardiovascular function, recipient preoperative plasma BNP (pBNP) levels emerged as a strong predictor. Elevated pBNP levels suggest impaired cardiac function, potentially reducing cardiac output and transplant kidney perfusion pressure, thereby increasing the risk of DGF [29]. This finding underscores the importance of comprehensive cardiovascular assessments for both donors and recipients during pre-transplant evaluations.

Interestingly, the study also identified several unconventional but clinically significant predictors. Microscopic hematuria (DuEC, DuRBC), which is well-established as a marker for chronic kidney disease progression, has been relatively understudied in transplantation but was found to be a valuable predictor for DGF [30]. Additionally, decreased donor APTT levels, indicating hypercoagulability, may increase the risk of microthrombosis, potentially exacerbating ischemia-reperfusion injury and impairing graft microcirculation [31]. This finding highlights the potential interaction between coagulation and immune responses in the development of DGF, although further experimental evidence is needed to validate this hypothesis [32]. Potassium ion (K^+) levels also demonstrated clinical relevance, as hypokalemia not only reflects simple electrolyte imbalance but may also indicate impaired tubular reabsorption and membrane instability, which could negatively impact early graft function recovery [33].

Further insights into the clinical timeline of DGF were provided by Cox regression and long-term prognosis analyses. Univariate Cox analysis revealed that donor type, recipient serum creatinine levels, and donor serum calcium levels significantly influenced DGF recovery time, highlighting the critical roles of organ quality and recipient baseline status in DGF duration [34]. Multivariate analysis identified preoperative dialysis duration as an independent predictor, revealing a strong association between prolonged dialysis burden and delayed DGF recovery [35]. This emphasizes the need for tailored management strategies for long-term dialysis patients in clinical practice.

Long-term prognosis analysis demonstrated the complex impacts of DGF on transplant outcomes. By categorizing patients into DGF Death, DGF Survival, Non-DGF Death, and Non-DGF Survival groups, several key findings emerged. The DGF Death Group exhibited the most severe renal dysfunction, with significantly higher serum creatinine ($\beta = 200.57$) and lower eGFR ($\beta = -39.91$), reflecting the poor prognosis associated with severe DGF. However, an important observation was that the DGF Survival Group showed renal function comparable to the Non-DGF Survival Group in most metrics, with only slightly elevated cystatin C levels ($\beta = 0.58$). This underscores that the occurrence of DGF does not necessarily lead to poor long-term outcomes, provided that early identification and timely interventions are implemented. Notably, comparisons between the DGF Survival and Death groups revealed no significant differences in DGF duration (16.66 ± 13.73 vs. 15.44 ± 14.62 days, $p = 0.735$), number of dialysis sessions (8.13 ± 7.39 vs. 7.78 ± 7.22, $p = 0.807$), or mean dialysis interval (1.89 ± 0.58 vs. 1.75 ± 0.70 days, $p = 0.649$). This unexpected finding suggests that these characteristics may not be decisive factors in determining long-term prognosis, which is more likely influenced by overall patient condition and perioperative management quality.

Longitudinal analysis showed significant declines in urea and cystatin C levels over time, supporting the notion that, even in the presence of DGF, most surviving patients can achieve gradual renal function improvement with appropriate management. These findings carry important clinical implications. Firstly, they emphasize the importance of early identification of high-risk patients. Secondly, they suggest that clinicians should not view DGF as a definitive indicator of poor prognosis but rather focus on early intervention and individualized treatment plans. Finally, these results provide transplant physicians with more detailed evidence for communicating long-term outcomes to patients [36].

Although this study has made significant progress in developing the first DGF prediction model based on multicenter data from China, several limitations require further exploration and refinement in future research. First, given the vast geographic diversity of China, regional differences in population characteristics (e.g., living environment, dietary habits) and medical practices may introduce heterogeneity, impacting the model's applicability. While this study used multicenter data, it primarily involved institutions from northern China. Expanding data collection to include more diverse regions and identifying population-specific factors will improve the model's generalizability. Second, the study spans a long data collection period, during which kidney transplantation techniques and perioperative management protocols may have evolved. Temporal effects may influence model accuracy and require additional attention in future research. Furthermore, ethical considerations in artificial intelligence,

particularly ensuring unbiased model application in clinical practice and enhancing interpretability, represent critical challenges. Incorporating novel biomarkers, such as genomic and transcriptomic data, into feature selection could further improve predictive power. Finally, prospective validation studies are essential for evaluating the model's effectiveness in real-world clinical applications, which represents a key direction for future research.

5 Conclusion

This study developed the first DGF prediction model based on multicenter data from China, integrating machine learning techniques with clinical data to achieve accurate post-transplant DGF risk prediction. In external validation, the LightGBM model demonstrated the best predictive performance (AUC = 0.80), providing a reliable decision-support tool for clinicians. SHAP analysis confirmed the importance of traditional risk factors such as donor eGFR and recipient preoperative plasma BNP levels while highlighting the potential value of novel predictors like microscopic hematuria and donor APTT. Cox regression analysis identified dialysis duration as a key factor influencing DGF recovery, while long-term follow-up results indicated that, although DGF increases early mortality risk, it does not significantly affect long-term outcomes in surviving patients. Importantly, DGF duration, number of dialysis sessions, and mean dialysis interval were not decisive factors for long-term prognosis. Based on these findings, the web-based prediction platform (http://www.kidney-dgf-match.cn/) was developed to assist clinicians in risk assessment and personalized treatment planning. This predictive model not only fills a research gap in the field of kidney transplantation in China but also provides new tools and insights for improving transplant outcomes.

Declarations

Acknowledge

We thank everyone in the department for their excellent work and selfless help.

Author contributions

PS and ZD conceptualized and designed the study. PS, XW, JL, KW, ZY, and WZ were responsible for data collection and curation. ZD performed the computational analysis, developed the machine learning models, and generated the figures. PS and ZD conducted the data analysis and interpretation. PS, ZD, and XW drafted the initial manuscript. HS, YC, LH, KM, and GW provided critical revision of the manuscript for important intellectual content and contributed to data collection from their respective centers. TG contributed to the study design. All authors reviewed and approved the final version of the manuscript. HS and YC supervised the project and provided administrative support.

Funding information

Central Government Support for Local Development 2023: Special Project for Construction of Scarce Disciplines in Strategic Emerging Industries - Experimental Surgery (Project No. 3110230005)

Data availability

To protect patient privacy and confidentiality, the raw data containing sensitive patient information used in this study are not publicly available. However, these data can be made available upon reasonable request to the first author or corresponding author, subject to appropriate data sharing agreements and ethical approvals.

The code used for data analysis and model development in this study is publicly accessible. All relevant scripts and code files can be found in our GitHub repository at https://github.com/Mr-SiSi/DGF-prediction-model-in-kidney-transplantation.

Ethics Statement

The study protocol was reviewed and approved by the Ethics Committee of the First Affiliated Hospital of China Medical University (approval number: AF-SOP-07-1.2-01). Due to the retrospective nature of this study, which involved the analysis of existing clinical data, the requirement for individual patient consent was waived by the ethics committee.

Conflict of Interest

The authors declare that the research was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

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Tables

Tables 2 and 3 are available in the Supplementary Files section.

Figures

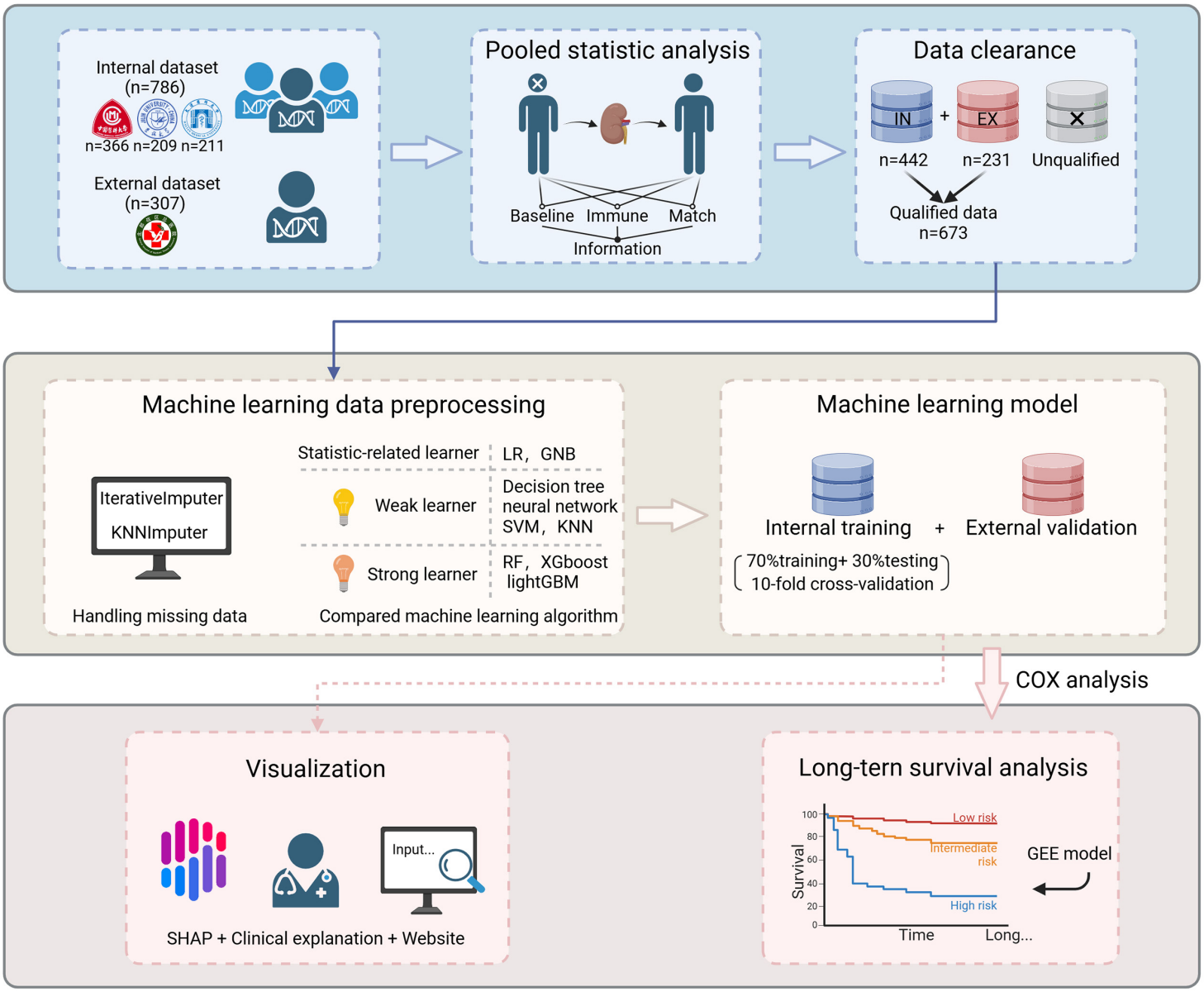


Figure 1

Study flowchart.

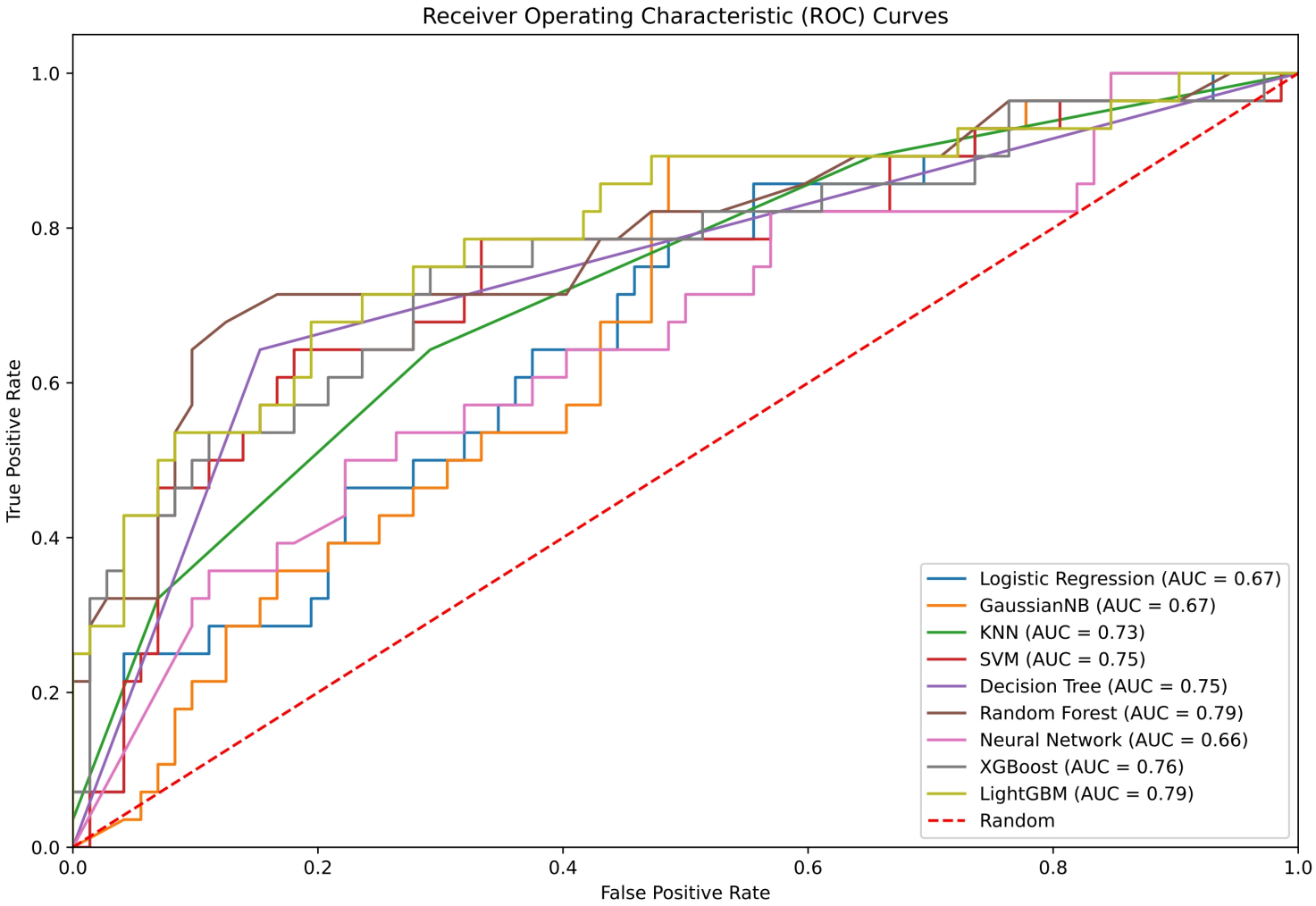


Figure 2

Comparison of Receiver Operating Characteristic (ROC) curves for nine machine learning models

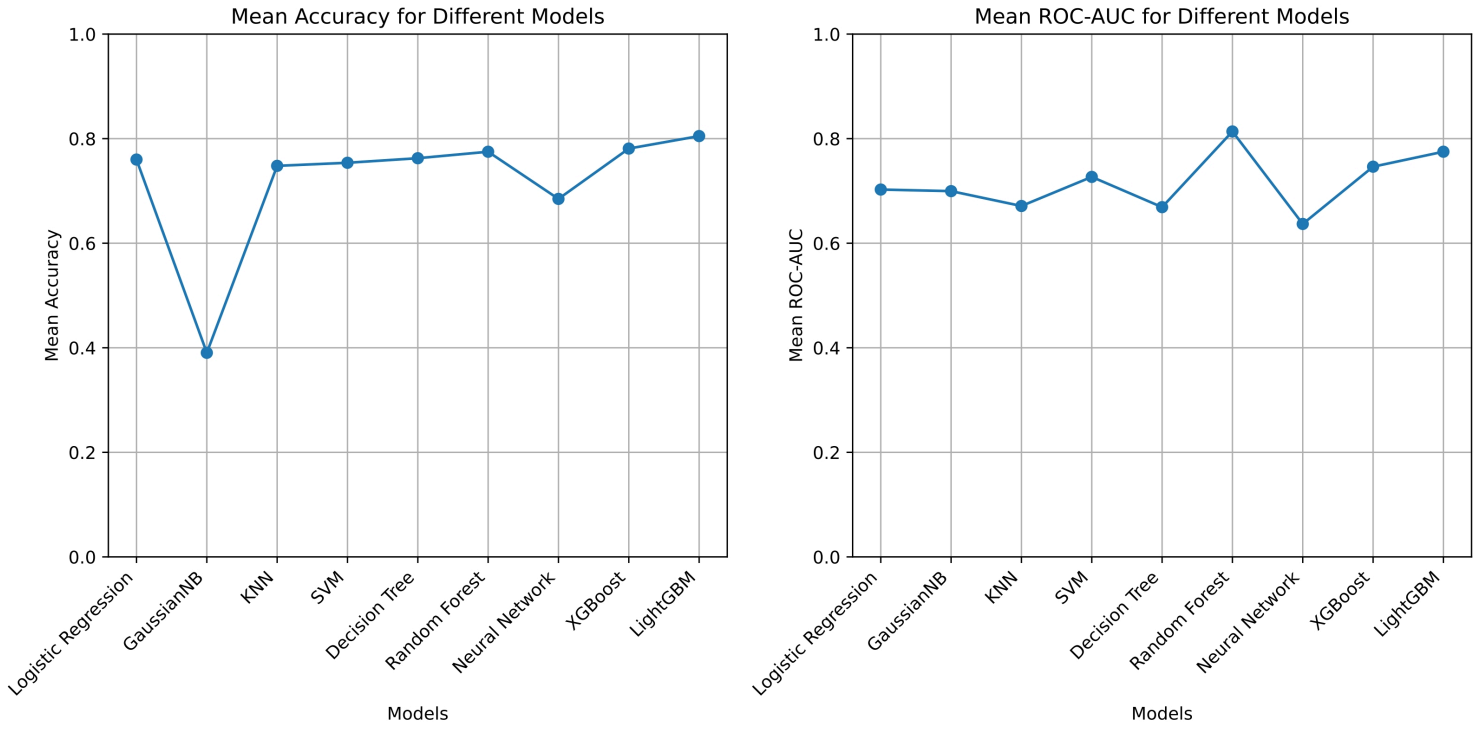


Figure 3

Line graph comparing the average accuracy and ROC-AUC values of nine machine learning models during 10-fold cross-validation

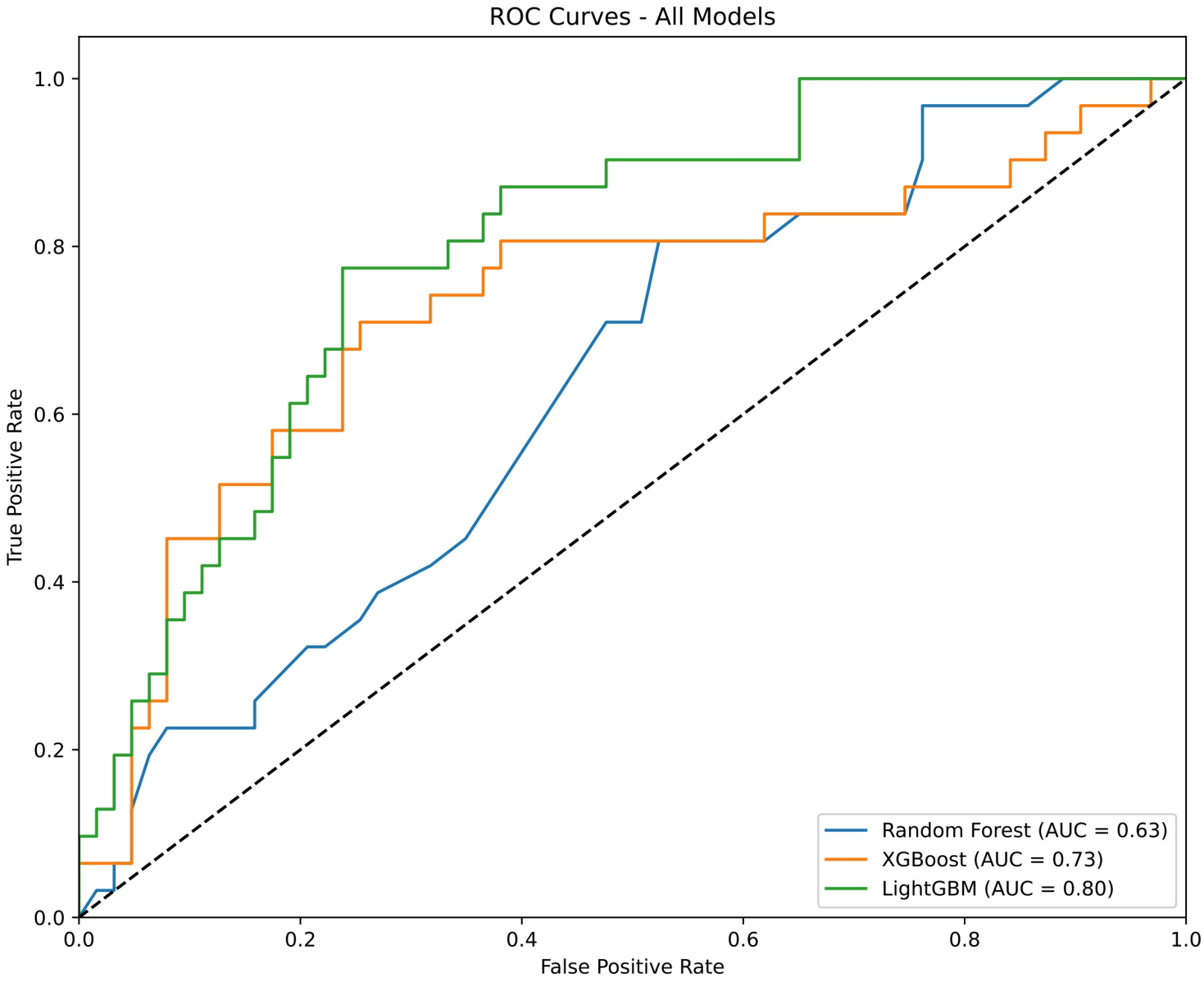


Figure 4

ROC Curves Comparison of Machine Learning Models on External Validation Dataset.

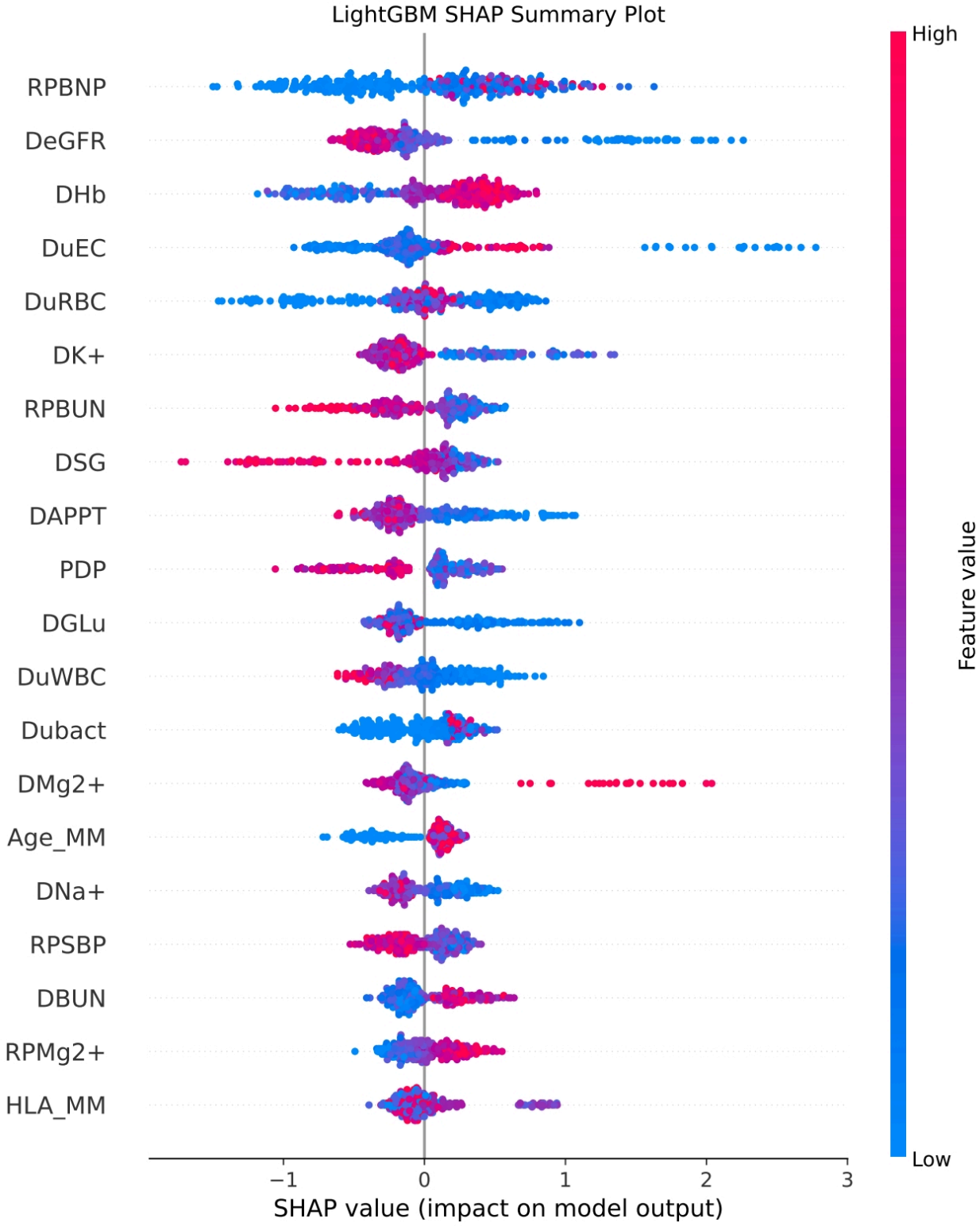


Figure 5

SHAP summary plot for the LightGBM model

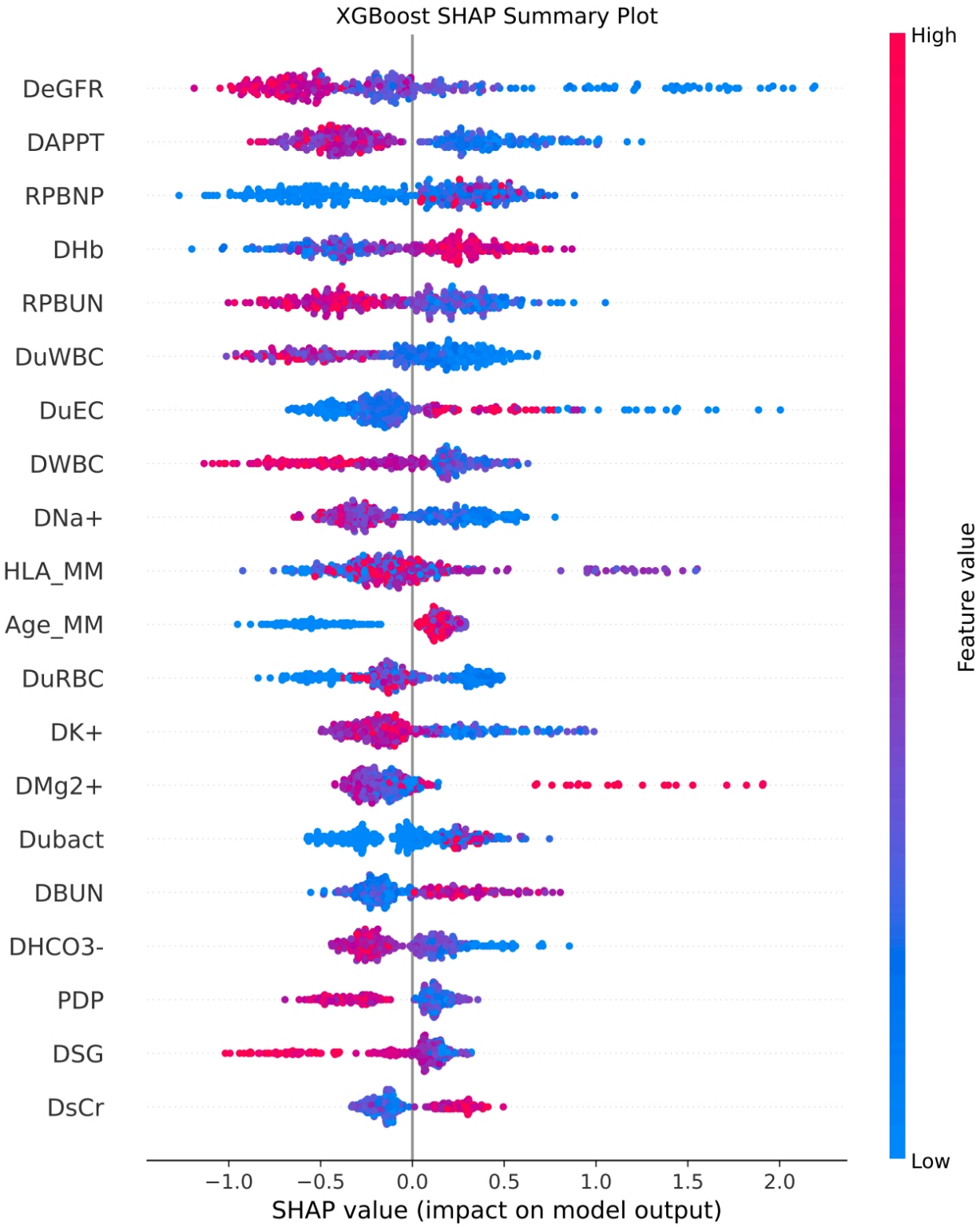


Figure 6

SHAP summary plot for the XGBoost model

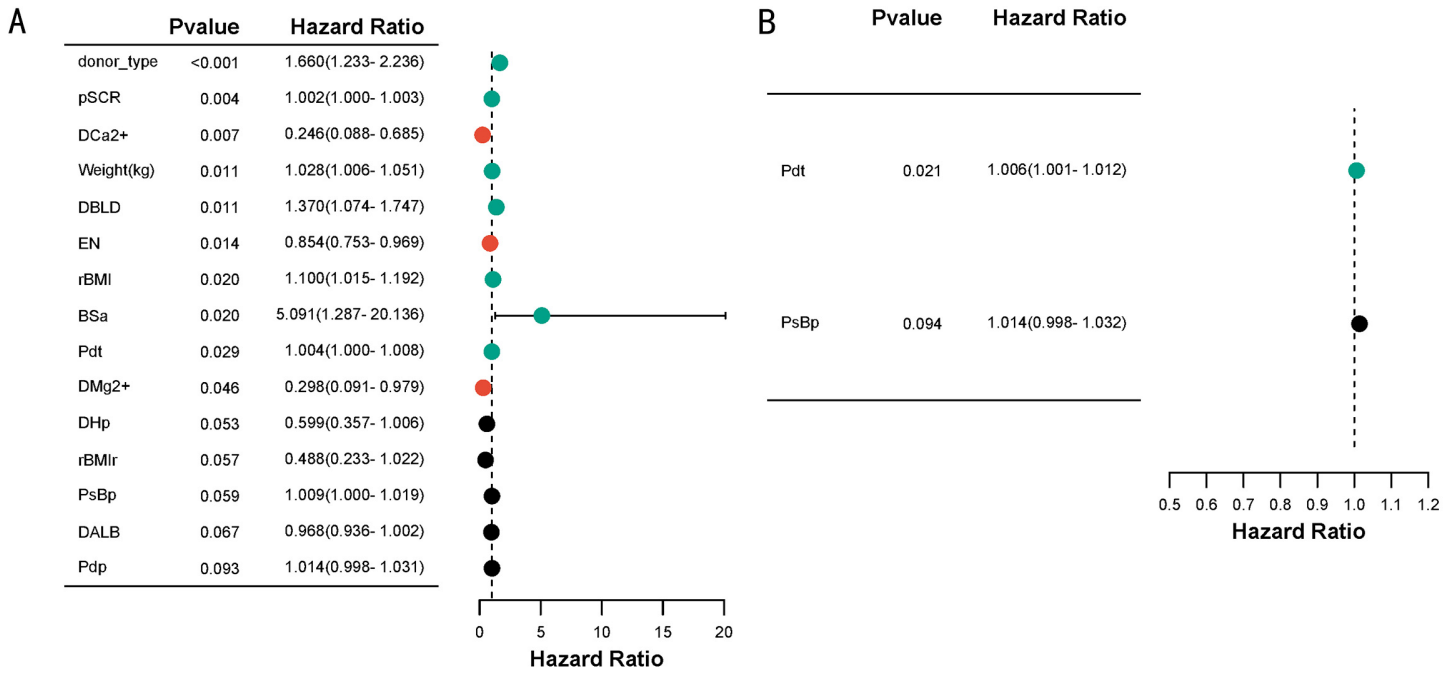


Figure 7

Univariate (A) and Multivariate (B) Cox regression analysis of factors associated with DGF recovery time.

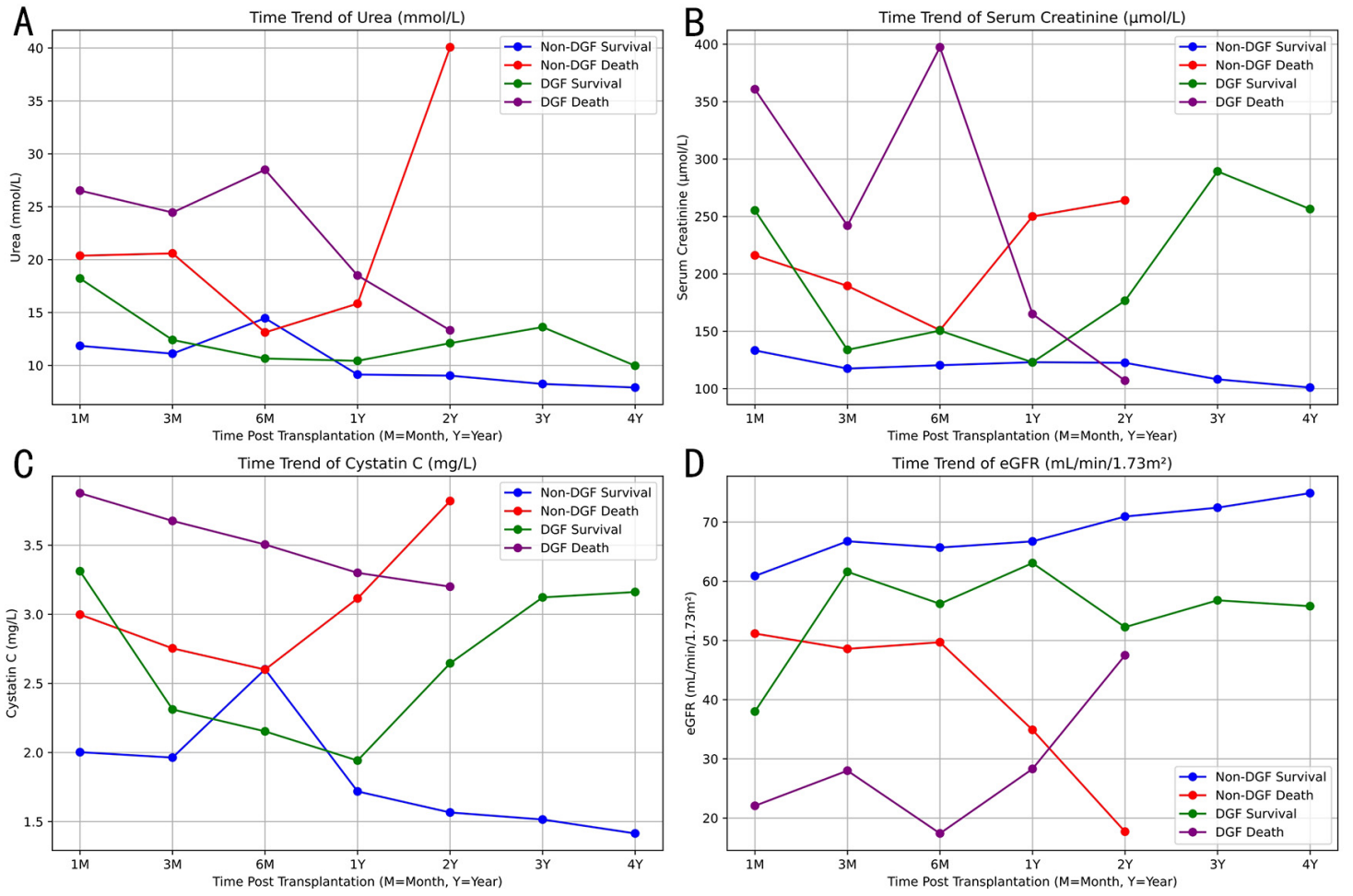


Figure 8

Longitudinal Changes in Renal Function Parameters Among Different Groups. A. Urea (mmol/L). B. Serum Creatinine (μmol/L). C. Cystatin C (mg/L). D. eGFR (mL/min/1.73m²).

Figure 9

Web Interface of the Online Risk Assessment Tool for DGF Prediction in Kidney Transplantation

Supplementary Files

This is a list of supplementary files associated with this preprint. Click to download.

- Tables.docx